Module 6: PHYSICS OF SENSORS:

1. How transducers differ from sensors? Explain concept of sensitivity, selectivity, accuracy and precision.
2. Define Hall Effect. Derive an expression of Hall Voltage. Explain physical significance of Hall coefficient.
3. Why photodiode is used in reversed bias for light sensing? Explain construction and working of Photodiode with I-V characteristics.
4. A sample of n type silicon has donar density of 10^{21}/m3. It is used in Hall effect experiment. If the sample of width 0.64 mm is kept in a magnetic field of 5500 Gauss with current density 400 A/m2. Calculate Hall potential, Hall Coefficient and Hall Angle (Given:Mobility is 0.17 m2/Vsec.)

Q1. How transducers differ from sensors? Explain concept of sensitivity, selectivity, accuracy and precision.

Ans:

Sensor: A sensor is a device that detects or senses a physical quantity (measurand) and converts it into another form of energy (usually electrical). The output may require further conditioning.

Transducer: A transducer is a device that converts a physical quantity from one form of energy to another form of energy. The output is generally electrical and is suitable for measurement, transmission or processing.

Difference:

Sensor	Transducer
1. Detects the measurand. 2. Output may not be directly in usable/electrical form. 3. Usually needs a transducer for conversion.	1. Converts the measurand from one form of energy to another. 2. Output is generally electrical and usable. 3. Completes the conversion process.

Key performance terms:

1. **Sensitivity:**
   It is the ratio of change in output to the corresponding change in input.

$$\text{Sensitivity (S)} = \frac{\Delta \text{Output}}{\Delta \text{Input}}$$

   Higher sensitivity means small change in input produces large change in output.

2. **Selectivity:**
   - It is the ability of a sensor to respond to the desired measurand and not to other influencing quantities (interfering variables).
   - A highly selective sensor is specific to the measurand.

3. **Accuracy:**
   - It is the closeness of measured value to the true or actual value.
   - Expressed in terms of % error.

$$\% \text{ Accuracy} = \left(1 - \frac{|\text{True value} - \text{Measured value}|}{\text{True value}}\right) \times 100$$

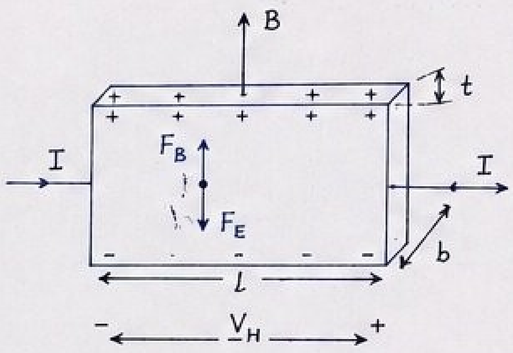
4. **Precision:**
   - It is the closeness of repeated measurements to each other under the same conditions.
   - High precision means low random error and good repeatability.

Q2. Define Hall Effect. Derive an expression of Hall Voltage. Explain physical significance of Hall coefficient.

Ans:

Hall Effect:

When a current carrying conductor or semiconductor is placed in a magnetic field perpendicular to the direction of current, a transverse electric field is produced across the conductor. This phenomenon is called Hall Effect and the transverse potential difference is called Hall Voltage.



Derivation:

Let current I flows along x-direction and magnetic field B along z-direction. Due to Lorentz force, charge carriers accumulate on the sides and a transverse electric field E_y is set up along y-direction. At equilibrium,

$$qE_y = qv_dB \quad \Rightarrow \quad E_y = v_dB$$

Hall Voltage, $V_H = E_y b = v_d B b \longrightarrow$ ...(1) ...(1)

Current, $I = nqAv_d = nq(bt)v_d \Rightarrow v_d = \frac{I}{nqbt}$ ...(2)

Substitute (2) in (1), $V_H = \left(\frac{I}{nqbt}\right)Bb = \frac{IB}{nqt}$

$$\boxed{V_H = \frac{IB}{nqt}}$$

Hall coefficient (R_H):

$$\boxed{R_H = \frac{E_y}{J_x B} = \frac{V_H t}{IB}}$$

For single type of charge,

$$R_H = \frac{1}{nq}$$

(+ve for holes, −ve for electrons)

Physical significance:

- R_H is a material constant.
- It gives information about type of charge carrier (sign) and their concentration (n).
- It is independent of dimensions (l, b, t) of the sample.

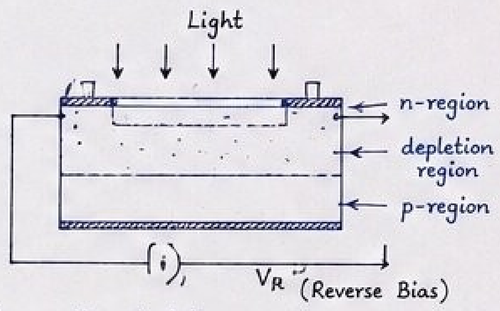
Q3. Why photodiode is used in reversed bias for light sensing? Explain construction and working of Photodiode with I-V characteristics.

Ans: Why reversed bias?

- Reversed bias widens the depletion region.
- It reduces junction capacitance and response time.
- It increases the electric field → carriers are quickly swept across → fast response.
- It reduces dark current and gives better linearity and sensitivity.

Construction:

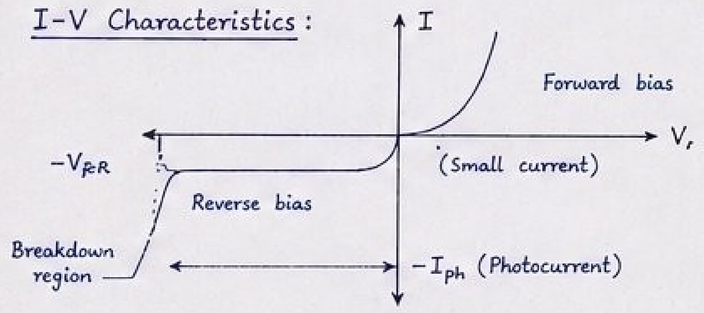
A photodiode is a p-n junction device with a window provided on the top surface to allow light to fall on the depletion region.



Working:

When light of energy $h\nu \geq E_g$ is incident on the depletion region, electron-hole pairs are generated. The built-in electric field in the depletion region separates them: electrons move to n-side and holes move to p-side. Thus a photocurrent flows in the external circuit.

I-V Characteristics:



In reverse bias, current ≈ constant photocurrent ($-I_{ph}$), almost independent of voltage until breakdown.

Q4. A sample of n type silicon has donor density of $10^{21}/m^3$. It is used in Hall effect experiment. If the sample of width 0.64 mm is kept in a magnetic field of 5500 Gauss with current density 400 A/m^2. Calculate Hall potential, Hall Coefficient and Hall Angle (Given: Mobility is 0.17 $m^2/Vsec.$).

Ans: Given:

Donor density, $n = 10^{21}/m^3$

Width, $b = 0.64\ mm = 0.64 \times 10^{-3}\ m$

Magnetic field, $B = 5500\ Gauss = 5500 \times 10^{-4}\ Tesla = 0.55\ T$

Current density, $J = 400\ A/m^2$

Mobility, $\mu = 0.17\ m^2/Vs$

(i) Hall potential, $V_H = R_H J B$

where Hall coefficient, $R_H = \frac{1}{nq}$

$q = 1.6 \times 10^{-19}\ C$

$\therefore R_H = \frac{1}{10^{21} \times 1.6 \times 10^{-19}} = 6.25 \times 10^{-3}\ m^3/C$

$V_H = (6.25 \times 10^{-3}) \times 400 \times 0.55$

$$\boxed{V_H = 1.375\ V}$$

(ii) Hall Coefficient,

$$R_H = \frac{1}{nq} = 6.25 \times 10^{-3}\ m^3/C$$

$$\boxed{R_H = 6.25 \times 10^{-3}\ m^3/C}$$

(iii) Hall Angle (θ_H),

$\tan\theta_H = \mu B$

$\theta_H = \tan^{-1}(\mu B)$

$= \tan^{-1}(0.17 \times 0.55)$

$= \tan^{-1}(0.0935)$

$$\boxed{\theta_H \approx 5.35^\circ}$$

$1\ Gauss = 10^{-4}\ Tesla$

$1\ mm = 10^{-3}\ m$